Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	28 June 2026
Team ID	STID-2026-7871
Project Name	India’s Agricultural Crop Production Analysis
Maximum Marks	4 Marks

Technical Architecture:

TECHNOLOGY STACK

The following technologies and tools were used to design, develop, analyze and deploy the project.

TECHNOLOGY / TOOL	PURPOSE	KEY FEATURES / USAGE
Tableau	Data Visualization and Dashboard Development	• Interactive Dashboards • Data Exploration • Storytelling • KPI & Calculated Fields • Filters & Parameters
Microsoft Excel	Data Storage and Initial Data Preparation	• Data Collection • Data Structuring • Basic Cleaning • Data Validation
Tableau Public	Publishing and Sharing Dashboards	• Publish Workbooks • Share Public Link • Make Visualizations Accessible
GitHub	Version Control and Collaboration	• Store Project Files • Track Changes • Team Collaboration
Microsoft Word	Documentation and Reporting	• Project Documentation • Report Preparation • Format & Structure
Internet / Web	Research and Data Collection	• Data Research • Validating Information • Reference and Insights

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	How user interacts with application e.g. Dashboards, stories, filters, charts etc.	Tableau Desktop(Interactive Dashboards)
2.	Application Logic-1	Logic for data preparation, data relationships, calculated fields, parameters and filters	Tableau Calculated Fields, Parameters, Sets, Groups, Filters
3.	Application Logic-2	Aggregation, measures, KPIs, trend analysis, comparative reports	Tableau Data Engine, Aggregations(SUM, AVG, COUNT, etc.)
4.	Application Logic-3	Story creation for presenting insights in a sequential and interactive manner.	Tableau Story, Story Points, Actions
5.	Database	Data Type, Configurations etc.	Microsoft Excel(.xlsx file)
6.	Cloud Database	Database Service on Cloud for publishing and sharing dashboards	Tableau Public(Cloud Platform)
7.	File Storage	File storage requirements for datasets, workbooks and reports	Local File System, Tableau Public Repository
8.	External API-1	Purpose of External API used in the application	Not used(Offline Dataset)
9.	External API-2	Purpose of External API used in the application	Not used
10.	Machine Learning Model	Purpose of Machine Learning Model	Not Applicable

11.	Infrastructure (Server / Cloud)	Application Deployment on Local System / Cloud Local Server Configuration:Windows OS Cloud Server Configuration : Tableau Public Cloud	Tableau Public, Cloud Infrastructure

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	List the open-source frameworks used	None(Tableau is a proprietary tool)
2.	Security Implementations	List all the security / access controls implemented, use of firewalls etc.	User authentication through Tableau Public account, Data privacy maintained, Access control via published links
3.	Scalable Architecture	Justify the scalability of architecture (3 – tier, Micro-services)	Scalable through Tableau Cloud, handles large datasets, supports multiple concurrent users on dashboards
4.	Availability	Justify the availability of application (e.g. use of load balancers, distributed servers etc.)	High availability through Public Cloud, accessible 24/7 over the web
5.	Performance	Design consideration for the performance of the application (number of requests per sec, use of Cache, use of CDN's) etc.	Optimized dashboards, extract data for fast performance, use of caching(Extracts), Tableau Cloud CDN for faster content delievery.